MILITARY SPECIFICATION

FILTER UNIT, CONTAMINATION ANALYSIS, GRAVIMETRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a filter assembly for the gravimetric determination of contaminant levels in automotive and aviation fuels, by the matched weight method.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 316 - Standard Test Method for Pore Size Characteristics of Membrane Filters for Use with Aerospace Fluids.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

HANDBOOK OF CHEMICAL MICROSCOPY

Volume 1. Third Edition

(Application for copies should be addressed to John Wiley and Sons, 605 Third Avenue, New York, NY 10016.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as specified herein.

3.1.1 Filter holder. The top and bottom sections of the filter holder shall be molded of transparent cellulose acetate-butyrate.

3.1.2 Filter membrane. The filter membrane shall be fabricated of pure, inert mixtures of cellulose acetate and cellulose nitrate.

3.1.3 Support pad. The support pad shall be fabricated of white cellulose.

3.2 Design and construction. The filter assembly shall consist of a transparent plastic filter holder, preassembled with two superimposed filters matched in weight to within 0.1 milligrams (mg) for particulate analysis of fluid samples by vacuum filtration. The filters shall be sealed between the holder sections with a cellulose pad under the filters to distribute sample flow evenly over the filter surface. The filter assembly shall meet the requirements specified herein and shall conform to the dimensions and shape specified in figure 1.

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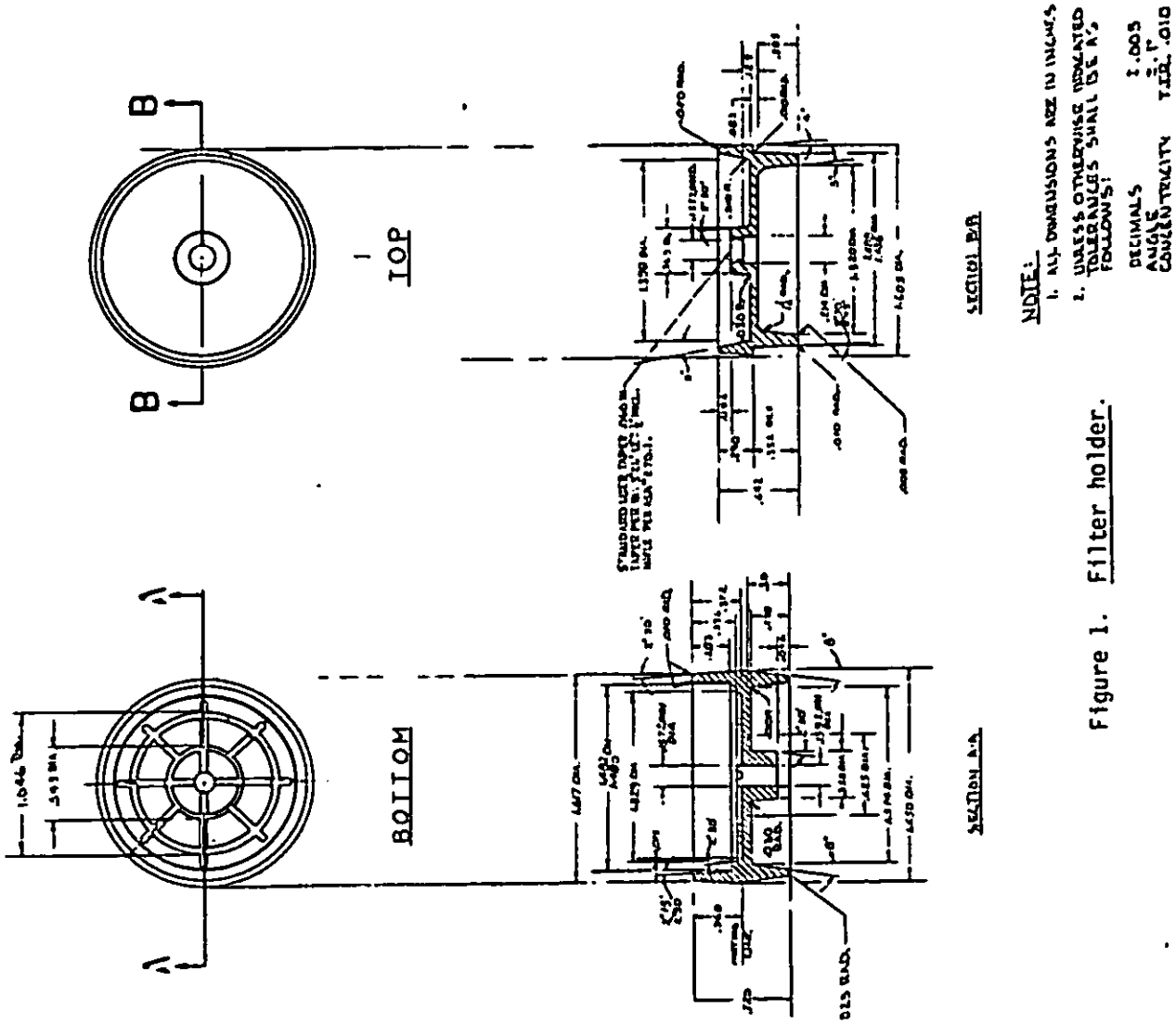


Figure 1. Filter holder.

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3.2.1 Filter.

3.2.1.1 Thickness. The filter disk shall have a thickness of 140 to 160 um (micrometers) when tested as specified in 4.4.1.

3.2.1.2 Diameter. The filter disk shall have a diameter of 37 ± 1/2 millimeter (mm).

3.2.1.3 Weight. The difference in weight between each filter disk of a matched pair shall be no more than 0.1 mg.

3.2.1.4 Pore size. The filter disk shall have a maximum pore size of 0.80 um when tested as specified in 4.4.2.

3.2.1.5 Porosity. The filter disk shall have a minimum porosity of 82 percent when tested as specified in 4.4.3.

3.2.1.6 Refractive index. The refractive index of the filter disk shall be 1.51 ± 0.01 when tested as specified in 4.4.4.

3.3 Performance.

3.3.1 Water extractables. Water extractables of the filter disk shall not exceed 4.0 percent when tested as specified in 4.4.5.

3.3.2 Flow rate. The water flow rate of the filter disk shall be as a minimum 157 milliliters/minute/cm^2 of filtration area with a differential pressure of 52 centimeters of mercury when tested as specified in 4.4.6.

3.3.3 Diffusibility. Water shall uniformly diffuse through the filter disk in 15 seconds, and there shall be no dry spots when tested as specified in 4.4.7.

3.3.4 Bubble point. The bubble point of the filter disk shall be 12 ± 2 psi (0.84 kg/cm^2) when tested as specified in 4.4.8.

3.4 Workmanship. Workmanship shall be first class throughout. The filter assembly shall be free from defects which detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one filter assembly.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Thickness	3.2.1.1	4.4.1	S-1	4.0
Pore size	3.2.1.4	4.4.2	S-1	4.0
Porosity	3.2.1.5	4.4.3	S-1	4.0
Refractive index	3.2.1.6	4.4.4	S-1	4.0
Water extractables	3.3.1	4.4.5	S-1	4.0
Flow rate	3.3.2	4.4.6	S-1	4.0
Diffusibility	3.3.3	4.4.7	S-1	4.0
Bubble point	3.3.4	4.4.8	S-1	4.0

4.3 Examination. The filter assembly shall be examined to determine compliance with all requirements specified herein. Examination shall be in accordance with, but shall not be limited to, the defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major:	
101	Component missing.
102	Filter not free of tears.
103	Filter not free of wrinkles.
104	Filter housing cracked.
105	Pad not free of tears.
106	Pad not free of wrinkles.

4.3.1 Packaging inspection. The inspection of the unit and intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Thickness. Thickness shall be determined by readings taken with a dial test indicator having a flat circular contact area mated to a flat anvil. The diameter of the contact surface shall be 3/8 to 5/8 inch (9.53 to 15.87 mm). The dial shall read accurately to 1 um (micrometer). The contact and anvil must be at "0" setting before and after each thickness reading. The contact shall not exert a pressure of more than 25 pounds per square inch (psi) (1.75 kg/cm^2) or less than

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1 psi (0.070 kg/cm^2). The filter shall be measured at the approximate four corner positions assuming an enclosed square, near, but not including the edge. One additional measurement shall be made at the center. The thickness shall meet the requirements of 3.2.1.1 (see 6.5).

4.4.2 Pore size. The maximum pore size shall be determined in accordance with ASTM F316.

4.4.3 Porosity. Porosity shall be determined by one of the following methods:

a. Calculate the theoretical weight of a solid filter by multiplying the volume by the density of the filter material. Determine the actual weight of the filter. Calculate the percent porosity by the following formula:

$$\text{Percent porosity} = \frac{\text{Theoretical weight of a solid filter} - \text{Actual weight of filter}}{\text{Theoretical weight of a solid filter.}} \times 100$$

b. Calculate the volume of the sample filter. Introduce mercury into the pores of the filter. Calculate the percent porosity by the following formula:

$$\text{Percent porosity} = \frac{\text{Volume of mercury introduced}}{\text{Theoretical volume of filter}} \times 100$$

The volume of mercury introduced into the pores of the filter shall be measured with an American Instrument Corp., Mercury Intrusion Tester or equal.

4.4.4 Refractive index. The refractive index of the filter disk shall be determined as follows:

Cut a 1 mm by 5 mm strip from a filter disk. Dimensions shall be approximate, and cutting shall be done with a sharp instrument so that smooth edges are obtained. Place the strip on a glass microscope slide. Add a drop of oil of known refractive index to the strip and cover the strip with a cover glass. Place the prepared slide on the stage of a microscope, having a magnification of 100X, and position so that the microscope is focused on an edge of the strip at the interface of the oil and the strip. If the refractive indices of the oil and the strip are different, a line called a Becke line will appear at the interface. Move the objective lens up or down so that the Becke line is moving from the strip to the oil. Upward movement of the lens indicates that the refractive index of the oil is greater than that of the strip. Downward movement of the lens indicates that the refractive index of the strip is greater than that of the oil. If the strip and Becke line cannot be brought into focus, the refractive index of the strip and the oil are the same. All oils used shall be within the tolerance of 3.2.1.6, and each test shall be performed with at least two oils, one of greater and one of lesser refractive index than the filter. (Theory of the refractive index test is given in the Handbook of Chemical Microscopy.)

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4.4.5 Water extractables. Weigh a filter disc to ± 0.0001 gr and record the weight. Heat the filter disk in an oven at 52°C (125.6°F) for 15 minutes. Remove the filter disk from the oven and place in a dessicator for one hour at room temperature. At the end of this period, weigh the filter disc to ± 0.0001 gram (gr). Heat 100 milliliters (ml) of distilled water in a beaker to boiling, drop the filter disk into the water and continue to boil for 30 minutes. Remove the filter disk from the water and dry in an oven at 52°C (125.6°F) for two hours. Dessicate for one hour at room temperature. Remove filter from dessicator and weigh to ± 0.0001 gr. Calculate percent extractables using the following formula:

$$\text{Percent extractables} = \frac{\text{Initial weight - final weight}}{\text{initial weight}} \times 100$$

4.4.6 Flow rate. Assemble laboratory apparatus capable of providing a 500 ml fluid reservoir, with a stopcock, at the input side of the filter assembly and a negative pressure of 52 centimeters of mercury (10 psi) at the outlet of the filter. Measure 500 ± 2 ml of filtered distilled water at 25°C (77°F) using a graduated beaker. Calculate the filtration area of the filter disk in square centimeters (cm^2). Float the filter disk on the surface of distilled water until the filter disk is completely wetted. Using forceps gently transfer the filter onto the pad in the filter holder and seal into position by closing the filter holder. Place the filter assembly into the laboratory testing apparatus making sure that the outlet side is at the negative pressure side of the apparatus. Close the stopcock of the reservoir and fill the reservoir with the 500 ± 2 ml of distilled water. Turn on the vacuum pump and open the reservoir stopcock. Record the time, within ± 0.2 seconds, for the 500 ml of water to pass through the filter. Calculate the flow rate as follows:

$$ml/min/cm^2 = \frac{60}{t\ (secs)} \times \frac{500}{\text{filtration area } (cm^2)}$$

4.4.7 Diffusibility. A dry filter shall be dropped flat onto the surface of distilled water and diffusion through the filter timed with a stopwatch. Wet areas will become darker while dry areas will not change in color intensity.

4.4.8 Bubble point. The bubble point test shall be conducted in accordance with ASTM-F316.

5. PACKAGING

5.1 Packaging. Packaging shall be level A, or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing forty-eight filter units, as specified, constitutes one unit of issue.

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5.1.2 Level A.

5.1.2.1 Unit packaging. Forty-eight filter units shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Six unit packages (288 filters) shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Forty-eight filter units shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Two intermediate packages (576 filters) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Two intermediate packages (576 filters) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged filters shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit, intermediate and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The filter unit covered by this specification is intended for the sampling of liquids from pressurized systems through analysis monitors for gravimetric determination of contaminant level.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item identification
6630-00-764-5761	FILTER UNIT, CONTAMINATION ANALYSIS, GRAVIMETRIC

6.4 The item specified herein is used with Sampler Assembly (NSN 6630-00-999-0573) and Sampling Adapter (NSN 6630-00-999-0754).

6.5 A suitable instrument is a dial test indicator made by the L.S. Starrett Co., consisting of base no. 654, head no. 656-261 reading in micrometers, and or no. 3 contact point.

Custodians:
Army - EA
Navy - MS
Air Force - 99

Preparing activity:
Defense Logistics Agency - DM

Project No. 6630-0278

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DOCUMENT IDENTIFIER (Number) AND TITLE
MIL-F-37706 FILTER UNIT, CONTAMINATION ANALYSIS, GRAVIMETRIC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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